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Rh-Catalyzed Carbonylative Cyclization of Propargylic Alcohols with Aryl Boronic Acids

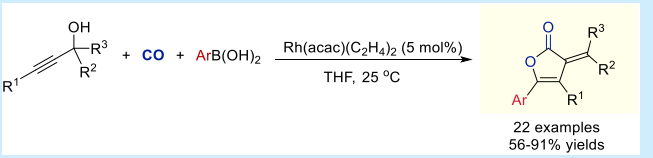
Wenxiang He, Wei-Feng Zheng, and Hui Qian*

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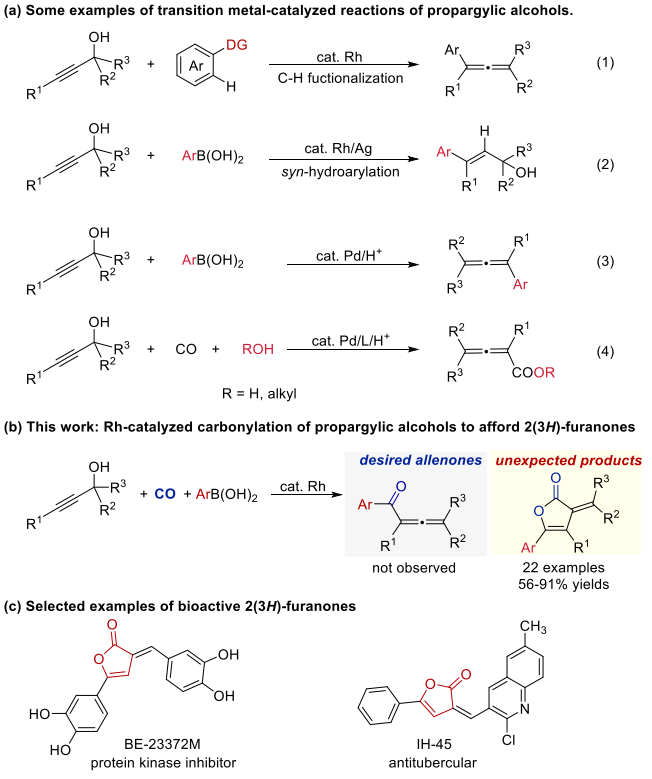
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ABSTRACT: 2(3*H*)-Furanones are tremendously important not only because of their wide occurrence in bioactive compounds but also due to their versatility in organic synthesis. Here, a straightforward approach to 2(3*H*)-furanones from readily available tertiary propargylic alcohols with arylboronic acids in the presence of CO using rhodium as a catalyst has been established. The method exhibits a broad substrate scope tolerating useful functional groups with a moderate to high stereoselectivity.



Butenolides represent privileged motifs commonly found in natural products, pharmaceuticals, and bioactive molecules.[1] Moreover, butenolides also serve as versatile building blocks in the total synthesis of natural products[2,3] and asymmetric catalysis.[4] Of particular interest, 2(3*H*)-furanones have also attracted considerable interest due to their potential in biological activity (Scheme 1c).[5] Thus, the development of new synthetic methods for 2(3*H*)-furanones is of high interest. On the other hand, propargylic alcohols are easily available and versatile building blocks in organic synthesis due to the presence of two functionalities.[6] Recently, our group have developed a series of transition-metal-catalyzed reactions of propargylic alcohols:[7,8] we reported that under the catalysis of Rh(III), the C−H functionalization of arenes with tertiary propargylic alcohols afforded tetra-substituted allenes (Scheme 1a, eq 1).[7a] In addition, Rh^{3+}/Ag^{+}-cocatalyzed reaction of propargylic alcohols with boronic acids would afford allylic alcohols (Scheme 1a, eq 2).[7b] However, the Pd/H^{+}-cocatalyzed reaction of propargylic alcohols with boronic acids afforded tri- or tetra-substituted allenes (Scheme 1a, eq 3).[7c] Furthermore, in the presence of CO and water or MeOH, with Pd and Brønsted acid as cocatalysts, propargylic alcohols would undergo carboxylation, affording 2,3-allenoic acids[7d,e] or allenoates[7f] (Scheme 1a, eq 4). We envisioned that trapping the allenylic rhodium intermediate in the presence of CO and boronic acids would result in the formation of not-readily available thermosensitive allenones under mild conditions (Scheme 1b).[9,10] However, 2(3*H*)-furanones were obtained unexpectedly. Here we wish to report this Rh-catalyzed carbonylation of tertiary propargylic alcohols affording 2(3*H*)-furanones exclusively with good to excellent stereoselectivity (Scheme 1b).

Scheme 1. Transition-Metal-Catalyzed Reactions of Propargylic Alcohols

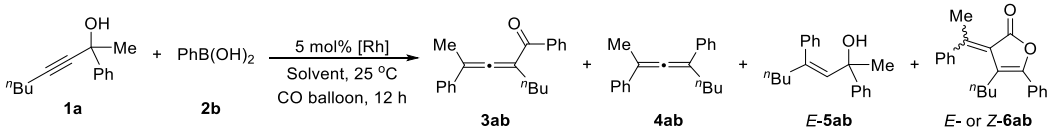


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Table 1. Optimization of the Reaction Conditions[a]



entry	[Rh]	Solvent	recovery of **1a** (%)	NMR yield of **3ab** (%)	NMR yield of **4ab** (%)	NMR yield of *E*-**5ab** (%)	NMR yield of *E*-**6ab** (%)	NMR yield of *Z*-**6ab** (%)
1	$[RhCl(COD)]_2$	THF	54	0	1	3	10	3
2	$[RhCl(C_2H_4)_2]_2$	THF	59	0	3	3	7	2
3	$[Rh(OH)(COD)]_2$	THF	23	0	5	18	44	6
4	$Rh(acac)(C_2H_4)_2$	THF	0	0	0	0	71	9
5	$Rh_2(OAc)_4$	THF	98	0	0	0	0	0
6	$[Cp^*RhCl_2]_2$	THF	87	0	0	0	0	0
7	$Rh(acac)(C_2H_4)_2$	Toluene	0	0	2	0	61	7
8	$Rh(acac)(C_2H_4)_2$	Dioxane	0	0	4	1	51	6
9	$Rh(acac)(C_2H_4)_2$	MeCN	41	0	2	6	12	4
10	$Rh(acac)(C_2H_4)_2$	MeOH	44	0	3	5	18	3
11	$Rh(acac)(C_2H_4)_2$	DCE	0	0	3	1	55	9

[a]Reaction conditions: **1a** (0.2 mmol), **2b** (0.4 mmol, 2.0 equiv), and Rh catalyst (5 mol% Rh) in solvent (1 mL) at 25 °C. The recoveries and NMR yields were determined by ^{1}H NMR analysis of the crude product using dibromomethane as the internal standard.

Initially, 2-phenyloct-3-yn-2-ol **1a** and phenylboronic acid **2b** were chosen as the model substrates using THF as the solvent with a CO balloon to test the hypothesis shown in Scheme 1b. However, the expected product allenone **3ab** was not detected when $[RhCl(COD)]_2$ was used as the catalyst. Interestingly, 1% yield of direct coupling product without CO incorporation allene **4ab**,[7c] 3% yield of allylic alcohol *E*-**5ab**,[7b] 10% yield of *E*-4-butyl-5-phenyl-3-(1-phenylethylidene)furan-2(3*H*)-one *E*-**6ab**, 3% yield of its stereoisomer *Z*-**6ab**, and 54% recovery of **1a** were detected (Table 1, entry 1). Different Rh catalysts were then examined. The yield of *E*-**6ab** slightly decreased when $[RhCl(C_2H_4)_2]_2$ was applied as the catalyst (Table 1, entry 2). By changing the catalyst to $[Rh(OH)(COD)]_2$, *E*-**6ab** was obtained in 44% yield (Table 1, entry 3). To our delight, when $Rh(acac)(C_2H_4)_2$ was employed, a 71% yield of *E*-**6ab** was obtained together with 9% of *Z*-**6ab** (Table 1, entry 4). We believe that the improvement is caused by the anion of the rhodium catalyst, where the enolate anion facilitates the transmetalation rather than the chloride anion. Among the screened Rh(II) and Rh(III) catalysts, $Rh_2(OAc)_4$ and $[Cp^*RhCl_2]_2$ did not yield the 2(3*H*)-furanone product *E*-**6ab**, with **1a** recovered with 98% and 87%, respectively (Table 1, entries 5 and 6). Solvents were found to be crucial for this transformation. The yield slightly decreased when toluene or dioxane was used as the solvent (Table 1, entries 7 and 8), while the efficiency decreased sharply when MeCN or MeOH was used (Table 1, entries 9 and 10). No improvement was obtained when DCE was employed (Table 1, entry 11). Among all of the solvents examined, THF was still the best. Thus, the optimal reaction conditions have been identified as shown in entry 4 of Table 1.

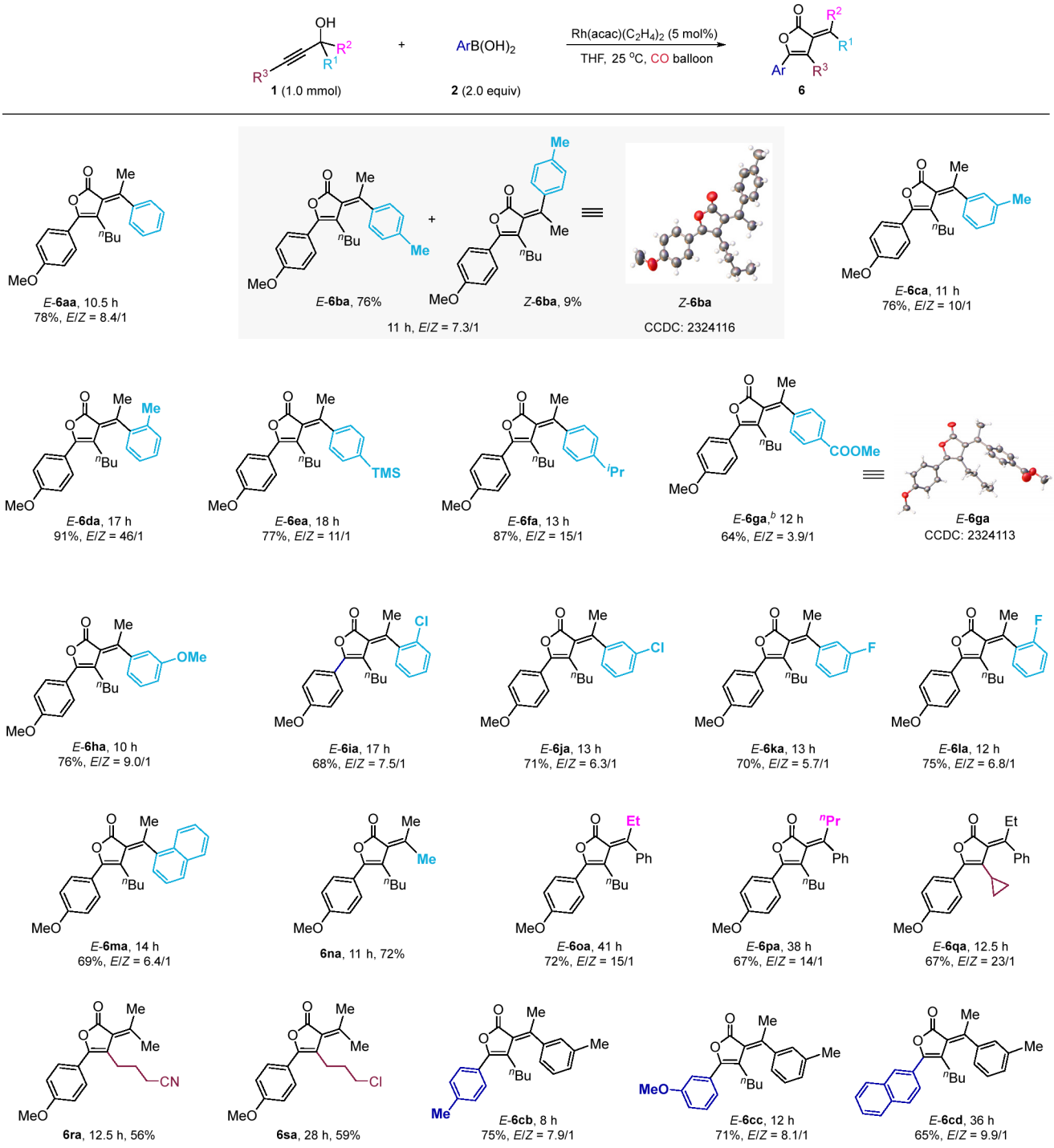
With the optimized reaction conditions in hand, we then explored the substrate scope (Scheme 2). A variety of tertiary propargylic alcohols with electron-rich, electron-neutral, and electron-deficient moieties substituted aryl groups (R^1) could be tolerated, affording 2(3*H*)-furanones in moderate to high yields. The stereoselectivity was established by single-crystal X-ray analysis of *Z*-**6ba** and *E*-**6ga**. The steric hindrance has an obvious influence on the stereoselectivity since the *E*/*Z* selectivity of **6da**, **6ca**, and **6ba** dropped in the order of *ortho*, *meta*, and *para*-substitution. Various functional groups with synthetic potentials, such as trimethylsilyl, ester, methoxyl, halide, cyano, etc., may be smoothly tolerated within the structure of propargylic alcohols for the reaction with arylboronic acid **2a** in the presence of CO to afford *E*-**6ea**, *E*-**6ga**, *E*-**6ha**, *E*-**6ia**, *E*-**6ja**, *E*-**6ka**, *E*-**6la**, and **6ra** in 56−77% isolated yields. Furthermore, 1-naphthyl was also tolerated (*E*-**6ma**). Apart from R^1 being an aryl group, the methyl group was also compatible, affording the desired 2(3*H*)-furanone **6na** in 72% yield. By changing R^2 to a longer alkyl chain such as ethyl or *n*-propyl, *E*-**6oa** and *E*-**6pa** were obtained in 72% and 67% yields, respectively, with a stereoselectivity higher than that of *E*-**6aa**. R^3 groups may be alkyl groups such as cyclopropyl, $-(CH_2)_3CN$, and $-(CH_2)_3Cl$, which worked smoothly to afford the corresponding products (*E*-**6qa**, **6ra**, and **6sa**). Besides, the reactivity of arylboronic acids was also investigated with **1c**, affording *E*-**6cb**, *E*-**6cc**, and *E*-**6cd** with a moderate stereoselectivity. We have also tried 3-furanboronic acid and 3-thiopheneboronic acid; however, the yield and stereoselectivity were not good (see the Supporting Information for more details).

A possible catalytic cycle is proposed as shown in Scheme 3.[11,12] An arylrhodium intermediate **B** formed by the transmetalation of Rh(I) species **A** with arylboronic acids would undergo CO insertion to form the acyl rhodium intermediate **C**. The OH-directed *syn*-insertion would result in the formation of alkenylrhodium intermediate **D**.[7a,b] The second CO insertion of the intermediate **D**, followed by an endomode insertion with the carbonyl group, would form intermediate **F**. We also conducted the experiment with ^{13}CO to confirm the double CO insertion process (see the Supporting Information for more details). Subsequent 1,3-allylic Rh migration would form intermediate **G** with the OH group at the *β*-position of the rhodium atom. Subsequently, *β*-OH elimination would then deliver the targeted products and regenerate the catalytically active Rh(I) species to finish the catalytic cycle. Another potential mechanism could involve the addition of arylcarbonyl to the Rh-carbonyl to form **F′**, followed by rearrangement of the carbon−rhodium

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Scheme 2. Substrate Scope Study[a]



[a]The reaction was conducted with 1.0 mmol of **1**, 2.0 mmol of **2**, and 5 mol% $Rh(acac)(C_2H_4)_2$ in 5 mL THF at 25 °C with a CO balloon unless otherwise noted. The ratios of *E*/*Z* stereoisomers were determined by ^1H NMR analysis of the crude product using dibromomethane as the internal standard. [b]10 mol% $Rh(acac)(C_2H_4)_2$ was used.

bond to yield a furan featuring a rhodium alkoxide **G′**, which was the enolate form of intermediate **G**.

In conclusion, we have developed the Rh-catalyzed carbonylative cyclization of tertiary propargylic alcohols with arylboronic acids in the presence of CO affording 2(3*H*)-furanones. The reaction proceeds via double CO insertion, and the mild reaction conditions enable a wide substrate scope tolerating many functional groups with a high stereoselectivity for the *exo*-C═C bond. Further studies related to this area are being pursued in our laboratory.

■ ASSOCIATED CONTENT

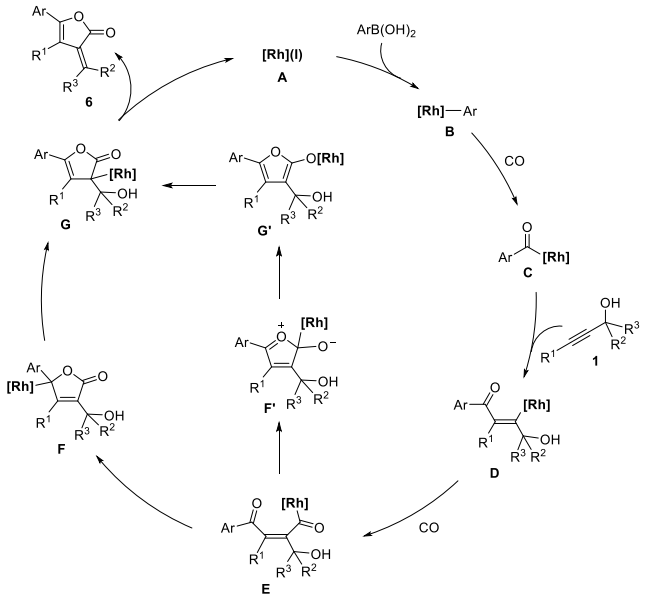
Data Availability Statement

The data underlying this study are available in the published article and its Supporting Information

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Scheme 3. Proposed Mechanism



Supporting Information

The Supporting Information is available free of charge at https://pubs.acs.org/doi/10.1021/acs.orglett.4c02364.

Optimizations, experimental procedures, and NMR spectra for obtained compounds (PDF)

Accession Codes

CCDC 2324113 and 2324116 contain the supplementary crystallographic data for this paper. These data can be obtained free of charge via www.ccdc.cam.ac.uk/data_request/cif, or by emailing data_request@ccdc.cam.ac.uk, or by contacting The Cambridge Crystallographic Data Centre, 12 Union Road, Cambridge CB2 1EZ, UK; fax: +44 1223 336033.

AUTHOR INFORMATION

Corresponding Author

Hui Qian − *Research Center for Molecular Recognition and Synthesis, Department of Chemistry, Fudan University, Shanghai 200433, P. R. China;* orcid.org/0000-0002-5606-9213; Email: qian_hui@fudan.edu.cn

Authors

Wenxiang He − *Research Center for Molecular Recognition and Synthesis, Department of Chemistry, Fudan University, Shanghai 200433, P. R. China*

Wei-Feng Zheng − *Research Center for Molecular Recognition and Synthesis, Department of Chemistry, Fudan University, Shanghai 200433, P. R. China*

Complete contact information is available at: https://pubs.acs.org/10.1021/acs.orglett.4c02364

Notes

The authors declare no competing financial interest.

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